Analysis 2 Final Test Solutions

1. Let $\{a_k\}$ and $\{b_k\}$ be real sequences such that $\sum_{k=1}^{\infty} a_k^2 < \infty$ and $\sum_{k=1}^{\infty} b_k^2 < \infty$. Show that $\sum_{k=1}^{\infty} a_k b_k$ converges absolutely.

Solution

Using that for all k we have

$$0 \leq (|a_k| - |b_k|)^2 = a_k^2 - 2|a_k b_k| + b_k^2$$

and so $2|a_k b_k| \leq a_k^2 + b_k^2$, we have that

$$\sum |a_k b_k| < \sum a_k^2 + \sum b_k^2 < \infty + \infty = \infty.$$

2. (a) Define the operator norm $\|T\|$ of a linear transformation $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.
   (b) Show that for all $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{x}\| = 1$, $\|T(\mathbf{x})\| \leq \|T\|$

Solution

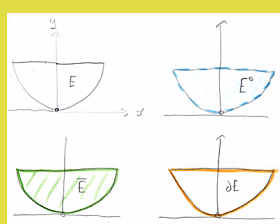
By definition $\|T\| = \sup_{\|\mathbf{x}\|>0} \frac{\|T(\mathbf{x})\|}{\|\mathbf{x}\|}$. So

$$\|T\| \geq \sup_{\|\mathbf{x}\|=1} \frac{\|T(\mathbf{x})\|}{\|\mathbf{x}\|} = \sup_{\|\mathbf{x}\|=1} \frac{\|T(\mathbf{x})\|}{1} = \sup_{\|\mathbf{x}\|=1} \|T(\mathbf{x})\|.$$

This inequality certaily holds when restricting to a particular value of $\mathbf{x}$.

3. Given that $E = \{(x, y) : y \geq x^2, 0 < y \leq 1\}$, sketch E^o, $\overline{E}$ and $\delta(E)$.

Solution



4. Suppose that $\mathbf{x}_k \to 0$ in $\mathbb{R}^n$ as $k \to \infty$, and that $\mathbf{y}_k$ is bounded. Prove that $\mathbf{x}_k \cdot \mathbf{y}_k \to 0$ as $k \to \infty$.

Solution

As $|\mathbf{x}_k \cdot \mathbf{y}_k| = \|\mathbf{x}_k\|\|\mathbf{y}_k\| \cos\theta_k \leq \|\mathbf{x}_k\|\|\mathbf{y}_k\|$ for all k and $\|\mathbf{y}_k\| < M$ for some M and all k, we have that

$$|\mathbf{x}_k \cdot \mathbf{y}_k| < \|\mathbf{x}_k\| M$$

for all k. So let $\varepsilon > 0$, as $\mathbf{x}_k \to 0$ there is some M such that $k \geq M$ impies that $\|\mathbf{x}_k\| < \varepsilon/M$ and so

$$\|\mathbf{x}_k \cdot \mathbf{y}_k\| < \|\mathbf{x}_k\| M < (\varepsilon/M)M = \varepsilon.$$

5. Suppose that K is compact in $\mathbb{R}^n$ and that for every $\mathbf{x} \in K$ there is an $r = r(\mathbf{x})$ such that $B_r(\mathbf{x}) \cap K = \{\mathbf{x}\}$. Prove that K is finite.

Solution

The family $\{B_r(\mathbf{x}) \cap K : \mathbf{x} \in K\}$ is an open cover of K, so as K is compact, has a finite subcover. But as each of these open sets is a singleton, a finite subcover consists of only a finite number of elements $\mathbf{x}$. So K is finite.

6. Prove whether of not the following limit exists

$$f(x, y) = \frac{\sin x \sin y}{x^2 + y^2} \text{ as } (x, y) \to (0, 0).$$

Solution

Taking the limit with respect to x for $y = 0$ we get that

$$\lim_{x \to 0} f(x, 0) = \lim_{x \to 0} \frac{0}{x^2 + 0} = 0.$$

Taking the limit as $x = y$ goes to 0 we get

$$\lim_{x \to 0} f(x, x) = \lim_{x \to 0} \frac{\sin^2(x)}{2x^2}.$$

Using l'hopital, this is

$$\lim_{x \to 0} \frac{2\sin(x)\cos(x)}{4x} = \lim_{x \to 0} \frac{\cos^2(x) - \sin^2(x)}{2} = 1/2.$$

As these path limits are different, the limit cannot exist.